Supermartingale Bounds for Multi-Phase Hypersonic Reachable Sets under Stochastic Forcing

Ethan Knox
Independent Researcher, Champaign, IL, USA

2026

Abstract

Keywords: stochastic reachability; Fokker-Planck equation; occupation measures; supermartingales; stochastic barrier certificates; concentration of measure; hybrid systems; atmospheric entry

MSC 2020:

Contents

Open items

1 Introduction

1.1 The price of hard bounds

1.2 Process noise and measurement noise

1.3 Thesis statement

1.4 Outline of contributions

2 The Stochastic Multi-Phase Model

2.1 Ito dynamics per mode

conditional on both prior papers (paper 1 for the embedding and the profile decomposition, paper 2 for the conditioning step).

2.2 Where the diffusion comes from

2.3 Guards under noise: hitting times and boundary behavior

2.4 The observation model with unbounded noise

3 From Liouville to Fokker-Planck

3.1 The stochastic occupation measure

3.2 The generator and the weak forward equation

3.3 Per-mode measures and stochastic linkage

4 Concentration Against Diffusion

4.1 Two competing limits

4.2 Diffusive spreading of the boundary layers

4.3 The critical noise scale

4.4 Where the real atmosphere sits

make-or-break condition for this paper: if measured density and wind variability put the real atmosphere above the critical noise scale, diffusion washes out the boundary layers and there is no skeleton left to bound around (paper 1's decomposition would not survive contact with noise).

4.5 Consequence for the certificate

5 Probabilistic Certificates

5.1 Why containment is unavailable

5.2 Supermartingales and the exit probability

5.3 Deriving the certificate

5.4 Numerical cross-check by sum-of-squares

5.5 Trading probability against volume

5.6 Set erosion and probabilistic tubes

6 Sequential Conditioning under Unbounded Noise

6.1 From restriction to reweighting

6.2 The conditioned stochastic occupation measure

6.3 Contraction in probability

6.4 Process noise against measurement information

7 Verification and Results

7.1 Recovering the deterministic limit

7.2 Where the real atmosphere sits

7.3 Exit probability against Monte Carlo

7.4 Probability against volume

7.5 Contraction under noisy returns

7.6 The three-way balance

7.7 Reproducibility environment

8 Discussion

8.1 Limitations

8.2 What the three papers establish together

8.3 Extensions

References

A Ito Calculus Across Guards

A.1 The generator of the hybrid diffusion

A.2 Hitting times and the boundary

A.3 The deterministic limit

B Diffusion Models from Atmospheric Data

B.1 Density variability

B.2 Wind variability

B.3 Polynomial form of the diffusion coefficient

C Supermartingale Inequalities

C.1 The maximal inequality

C.2 Finite-horizon and stopped versions

C.3 Composition over epochs

D Proof of the Concentration-Diffusion Threshold

D.1 Widths of the two effects

D.2 The threshold

D.3 The inflated residual